The Coh Category

Verifier-Certified Trace Composition

Version 1.0 - Operational Foundation

Core thesis. A transition is not a morphism merely because two states can be named. In Coh, a transition becomes a morphism only after deterministic verifier acceptance and satisfaction of the resource law

$$V(y) + \mathrm{Spend}(\tau) \leq V(x) + \mathrm{Defect}(\tau).$$

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Prepared as a self-contained V1 academic foundation for the Coh category.

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1 Introduction

In ordinary category-theoretic modeling, morphisms are typically part of the starting data. One declares objects, declares arrows between them, and then studies how those arrows compose. That order is often appropriate for abstract algebra, topology, and semantics. It is less appropriate for verification-sensitive computation, where the existence of a syntactic transition does not imply that the transition should be allowed.

The Coh viewpoint reverses the order of construction:

$$\text{proposal} \quad \longrightarrow \quad \text{verification} \quad \longrightarrow \quad \text{morphism}.$$

A transition is first proposed as a finite trace. It becomes a morphism only if it passes a deterministic verifier and satisfies the governing resource inequality

$$V(y) + \mathrm{Spend}(\tau) \leq V(x) + \mathrm{Defect}(\tau). \tag{1}$$

Thus morphisms are not freely given and later decorated by costs. They are *earned* by certification.

1.1 Purpose of V1

This paper gives the V1 foundation of the Coh category. It focuses only on the operational layer:

(i) finite traces between observable states;
(ii) deterministic verifier acceptance;
(iii) exact additive bookkeeping for spend and defect;
(iv) the category theorem for admissible traces;
(v) the global telescoping resource law.

The later projection-aware layer is not needed to prove the V1 category theorem. In later versions, hidden realizations project to observable traces and induce a semantic envelope defect. That extension is structurally compatible with the V1 paper, but V1 is intentionally kept minimal so that the categorical foundation is clean.

1.2 Contributions

The paper proves the following statements from explicit assumptions.

(1) **Admissibility generates hom-sets.** A hom-set is the set of traces satisfying verifier acceptance and the Coh resource law.
(2) **Certified traces compose.** Under verifier closure and additive bookkeeping, concatenation preserves admissibility.
(3) **Certified traces form a category.** Empty traces serve as identities, concatenation is associative, and certified closure gives composition.
(4) **Local resource laws telescope.** A finite admissible trajectory obeys a global inequality bounding terminal valuation plus accumulated spend by initial valuation plus accumulated defect.

1.3 Relation to existing categorical structures

Lawvere metric spaces and enriched categories allow costs or distances to decorate categorical structure. The Coh construction differs in its order of generation. The trace does not first exist as a morphism with a cost annotation. Instead, the trace is admitted as a morphism only after passing a verifier and satisfying the resource inequality. This makes Coh closer in spirit to proof-carrying execution and certified transition systems, but expressed through a categorical composition theorem.

2 Operational Data

2.1 State space and valuation

Definition 2.1 (Operational state space)**.** An operational state space is a set X. Elements of X are called observable states.

Definition 2.2 (Valuation)**.** A valuation is a function

$$V : X \to \mathbb{Q}_{\geq 0},$$

where $V(x)$ measures the current risk, violation, unresolved obligation, or remaining potential associated with state x.

Remark 2.3. The codomain $\mathbb{Q}_{\geq 0}$ is chosen for exact verifier-facing arithmetic. A real-valued variant is mathematically natural, but the rational codomain better matches deterministic implementation boundaries. Nothing in the V1 category theorem requires completeness of the real numbers.

2.2 Primitive steps and traces

Definition 2.4 (Primitive step)**.** A primitive step $\sigma : x \rightsquigarrow y$ consists of a source state $x \in X$, a target state $y \in X$, and step-level bookkeeping values

$$s(\sigma), d(\sigma) \in \mathbb{Q}_{\geq 0}.$$

The number $s(\sigma)$ is the spend of the step and $d(\sigma)$ is its defect allowance.

Definition 2.5 (Finite trace)**.** A finite trace $\tau : x \rightsquigarrow y$ is a finite composable sequence

$$x = x_0 \xrightarrow{\sigma_1} x_1 \xrightarrow{\sigma_2} \cdots \xrightarrow{\sigma_n} x_n = y.$$

The length of τ is n. The empty trace at x is denoted $\varepsilon_x : x \rightsquigarrow x$ and has length zero.

Definition 2.6 (Trace concatenation)**.** If $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$ are traces, their concatenation is

$$\tau_2 * \tau_1 : x \rightsquigarrow z,$$

obtained by appending the step sequence of τ_2 after the step sequence of τ_1.

Definition 2.7 (Additive bookkeeping)**.** For a trace

$$\tau = (\sigma_1, \ldots, \sigma_n),$$

define

$$\mathrm{Spend}(\tau) = \sum_{i=1}^{n} s(\sigma_i), \qquad \mathrm{Defect}(\tau) = \sum_{i=1}^{n} d(\sigma_i). \tag{2}$$

For the empty trace, both sums are zero.

Lemma 2.8 (Bookkeeping additivity)**.** *For composable traces τ_1 and τ_2,*

$$\mathrm{Spend}(\tau_2 * \tau_1) = \mathrm{Spend}(\tau_1) + \mathrm{Spend}(\tau_2),$$

and

$$\mathrm{Defect}(\tau_2 * \tau_1) = \mathrm{Defect}(\tau_1) + \mathrm{Defect}(\tau_2).$$

Proof. Both identities follow directly from the definition of concatenation and finite sums over appended lists. □

3 Verifier Structure

Definition 3.1 (Deterministic verifier)**.** A deterministic verifier is a function

$$\mathrm{RV} : \mathrm{Trace}(X) \to \{\mathrm{ACCEPT}, \mathrm{REJECT}\}.$$

A trace τ is verifier-accepted when $\mathrm{RV}(\tau) = \mathrm{ACCEPT}$.

Axiom 3.2 (Identity acceptance)**.** For every state $x \in X$,

$$\mathrm{RV}(\varepsilon_x) = \mathrm{ACCEPT}\,.$$

Axiom 3.3 (Verifier closure under concatenation)**.** If $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$ satisfy

$$\mathrm{RV}(\tau_1) = \mathrm{ACCEPT}, \qquad \mathrm{RV}(\tau_2) = \mathrm{ACCEPT},$$

then

$$\mathrm{RV}(\tau_2 * \tau_1) = \mathrm{ACCEPT}\,.$$

Remark 3.4 (Why closure is an axiom)*.* Verifier closure is not automatic. A verifier may accept two traces individually while rejecting their composite because of history, rate limits, or context-sensitive constraints. V1 isolates the exact condition needed for a category theorem: the accepted traces must be closed under compatible concatenation.

4 Admissibility

Definition 4.1 (Coh admissibility)**.** A trace $\tau : x \rightsquigarrow y$ is *Coh-admissible* if both conditions hold:

(i) $\mathrm{RV}(\tau) = \mathrm{ACCEPT}$;

(ii) the governing inequality holds:

$$V(y) + \mathrm{Spend}(\tau) \le V(x) + \mathrm{Defect}(\tau). \tag{3}$$

Remark 4.2 (Spend and defect are separated)*.* The two channels in (3) have distinct meanings. Spend is cost paid by the transition. Defect is admissible slack, uncertainty, tolerance, or unresolved error. Collapsing them into one scalar would erase the distinction between paid cost and tolerated deviation.

Lemma 4.3 (Identity admissibility)**.** *For every state $x \in X$, the empty trace ε_x is Coh-admissible.*

Proof. By Axiom 3.2, $\mathrm{RV}(\varepsilon_x) = \mathrm{ACCEPT}$. Since the empty trace has zero spend and zero defect,

$$V(x) + \mathrm{Spend}(\varepsilon_x) = V(x) + 0 = V(x) + 0 = V(x) + \mathrm{Defect}(\varepsilon_x).$$

Thus the governing inequality holds with equality. □

Lemma 4.4 (Closure of admissibility)**.** *Let $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$ be Coh-admissible traces. Then $\tau_2 * \tau_1 : x \rightsquigarrow z$ is Coh-admissible.*

Proof. Verifier acceptance of $\tau_2 * \tau_1$ follows from Axiom 3.3. It remains to prove the resource inequality.

Since τ_1 and τ_2 are admissible,

$$V(y) + \mathrm{Spend}(\tau_1) \le V(x) + \mathrm{Defect}(\tau_1), \tag{4}$$

$$V(z) + \mathrm{Spend}(\tau_2) \le V(y) + \mathrm{Defect}(\tau_2). \tag{5}$$

Adding (4) and (5) gives

$$V(y) + V(z) + \mathrm{Spend}(\tau_1) + \mathrm{Spend}(\tau_2) \le V(x) + V(y) + \mathrm{Defect}(\tau_1) + \mathrm{Defect}(\tau_2).$$

Cancellation of the finite term $V(y)$ yields

$$V(z) + \mathrm{Spend}(\tau_1) + \mathrm{Spend}(\tau_2) \le V(x) + \mathrm{Defect}(\tau_1) + \mathrm{Defect}(\tau_2).$$

By Lemma 2.8, this becomes

$$V(z) + \mathrm{Spend}(\tau_2 * \tau_1) \leq V(x) + \mathrm{Defect}(\tau_2 * \tau_1).$$

Hence $\tau_2 * \tau_1$ is admissible. □

5 Construction of the Coh Category

Definition 5.1 (Coh hom-set)**.** For states $x, y \in X$, define

$$\mathrm{Hom}_{\mathrm{Coh}(S)}(x, y) = \{\tau : x \rightsquigarrow y \mid \tau \text{ is Coh-admissible}\}.$$

Definition 5.2 (Identity morphism)**.** For each $x \in X$, define

$$\mathrm{id}_x := \varepsilon_x.$$

Definition 5.3 (Composition)**.** For $\tau_1 \in \mathrm{Hom}(x, y)$ and $\tau_2 \in \mathrm{Hom}(y, z)$, define

$$\tau_2 \circ \tau_1 := \tau_2 * \tau_1.$$

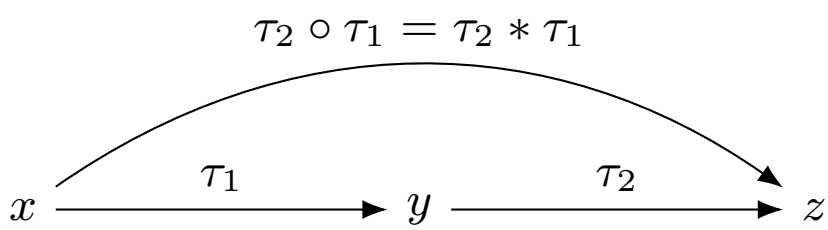

Figure 1. Certified composition. If $\tau_1 : x \rightsquigarrow y$ and $\tau_2 : y \rightsquigarrow z$ are admissible, their composite is trace concatenation.

Theorem 5.4 (Coh category)**.** *Let $S = (X, V, \mathrm{Spend}, \mathrm{Defect}, \mathrm{RV})$ satisfy identity acceptance, verifier closure under concatenation, and additive trace bookkeeping. Then the data above form a category, denoted* $\mathrm{Coh}(S)$.

Proof. We verify the category axioms.

Identities. By Lemma 4.3, ε_x is admissible for every $x \in X$. Hence $\mathrm{id}_x \in \mathrm{Hom}(x, x)$.

Closure of composition. If $\tau_1 \in \mathrm{Hom}(x, y)$ and $\tau_2 \in \mathrm{Hom}(y, z)$, then τ_1 and τ_2 are admissible. By Lemma 4.4, $\tau_2 * \tau_1$ is admissible, so $\tau_2 \circ \tau_1 \in \mathrm{Hom}(x, z)$.

Associativity. Let $\tau_1 : x \rightsquigarrow y$, $\tau_2 : y \rightsquigarrow z$, and $\tau_3 : z \rightsquigarrow w$ be admissible. Composition is concatenation of finite lists of primitive steps. List concatenation is associative, so

$$(\tau_3 \circ \tau_2) \circ \tau_1 = (\tau_3 * \tau_2) * \tau_1 = \tau_3 * (\tau_2 * \tau_1) = \tau_3 \circ (\tau_2 \circ \tau_1).$$

Unit laws. For any admissible $\tau : x \rightsquigarrow y$,

$$\varepsilon_y * \tau = \tau, \qquad \tau * \varepsilon_x = \tau,$$

because the empty trace is the empty list of steps. Therefore

$$\mathrm{id}_y \circ \tau = \tau, \qquad \tau \circ \mathrm{id}_x = \tau.$$

Thus all category axioms hold. □

6 Trace Algebra and Global Budget Law

The category theorem shows that admissible traces compose. The next result explains what composition does to the resource inequality over long trajectories.

Definition 6.1 (Admissible trajectory)**.** A finite trajectory

$$x_0 \xrightarrow{r_0} x_1 \xrightarrow{r_1} \cdots \xrightarrow{r_{N-1}} x_N$$

is admissible when each step $r_i : x_i \rightsquigarrow x_{i+1}$ is Coh-admissible. Write

$$S_i := \mathrm{Spend}(r_i), \qquad D_i := \mathrm{Defect}(r_i).$$

Lemma 6.2 (One-step growth inequality)**.** *For every admissible step $r_i : x_i \rightsquigarrow x_{i+1}$,*

$$V(x_{i+1}) - V(x_i) \le D_i - S_i.$$

Proof. Starting from admissibility,

$$V(x_{i+1}) + S_i \le V(x_i) + D_i.$$

Subtracting $V(x_i) + S_i$ from both sides gives the claim. □

Theorem 6.3 (Telescoping resource law)**.** *For every finite admissible trajectory*

$$x_0 \to x_1 \to \cdots \to x_N,$$

one has

$$V(x_N) + \sum_{i=0}^{N-1} S_i \le V(x_0) + \sum_{i=0}^{N-1} D_i. \tag{6}$$

Proof. By admissibility of each step,

$$S_i \le V(x_i) - V(x_{i+1}) + D_i.$$

Summing from $i = 0$ to $N - 1$ gives

$$\sum_{i=0}^{N-1} S_i \le \sum_{i=0}^{N-1} \big(V(x_i) - V(x_{i+1})\big) + \sum_{i=0}^{N-1} D_i.$$

The valuation sum telescopes:

$$\sum_{i=0}^{N-1} \big(V(x_i) - V(x_{i+1})\big) = V(x_0) - V(x_N).$$

Therefore

$$\sum_{i=0}^{N-1} S_i \le V(x_0) - V(x_N) + \sum_{i=0}^{N-1} D_i,$$

which is equivalent to (6). □

Corollary 6.4 (Risk bound)**.** *If spend is nonnegative, then*

$$V(x_N) \le V(x_0) + \sum_{i=0}^{N-1} D_i.$$

Proof. Drop the nonnegative term $\sum_i S_i$ from the left side of (6). □

Corollary 6.5 (Spend budget)**.** *If $V \ge 0$, then*

$$\sum_{i=0}^{N-1} S_i \le V(x_0) + \sum_{i=0}^{N-1} D_i.$$

Proof. From Theorem 6.3,

$$\sum_i S_i \leq V(x_0) - V(x_N) + \sum_i D_i.$$

Since $V(x_N) \geq 0$, the result follows. □

7 Worked Example

The following finite system shows how the inequality generates admissible morphisms and how composition preserves certification.

Example 7.1 (Three-state descent)**.** Let

$$X = \{A, B, C\}, \qquad V(A) = 5, \quad V(B) = 4, \quad V(C) = 3.$$

Let $\sigma_1 : A \rightsquigarrow B$ and $\sigma_2 : B \rightsquigarrow C$ be primitive steps with

$$s(\sigma_1) = 1, \quad d(\sigma_1) = 0, \qquad s(\sigma_2) = 1, \quad d(\sigma_2) = 0.$$

Assume the verifier accepts both steps.

Proposition 7.2. *The primitive traces σ_1 and σ_2 are Coh-admissible.*

Proof. For σ_1,

$$V(B) + s(\sigma_1) = 4 + 1 = 5 = 5 + 0 = V(A) + d(\sigma_1).$$

For σ_2,

$$V(C) + s(\sigma_2) = 3 + 1 = 4 = 4 + 0 = V(B) + d(\sigma_2).$$

Both satisfy the resource inequality with equality, and both are verifier-accepted by assumption. □

Proposition 7.3. *The composite trace $\tau = \sigma_2 * \sigma_1 : A \rightsquigarrow C$ is Coh-admissible.*

Proof. The verifier accepts the composite by verifier closure. The spend and defect are

$$\mathrm{Spend}(\tau) = 1 + 1 = 2, \qquad \mathrm{Defect}(\tau) = 0 + 0 = 0.$$

Thus

$$V(C) + \mathrm{Spend}(\tau) = 3 + 2 = 5 = V(A) + \mathrm{Defect}(\tau).$$

So the composite is admissible. □

Remark 7.4 (Boundary behavior)*.* This example sits exactly on the boundary of admissibility. Every step spends all valuation decrease available to it, and no defect is used. Such equality cases are useful for testing implementations because any arithmetic or ordering error can flip acceptance.

8 Figures and Operational Interpretation

Figure 2 summarizes the V1 execution wedge. Candidate traces are proposed and passed to the verifier. Only accepted traces satisfying the resource law enter the categorical layer.

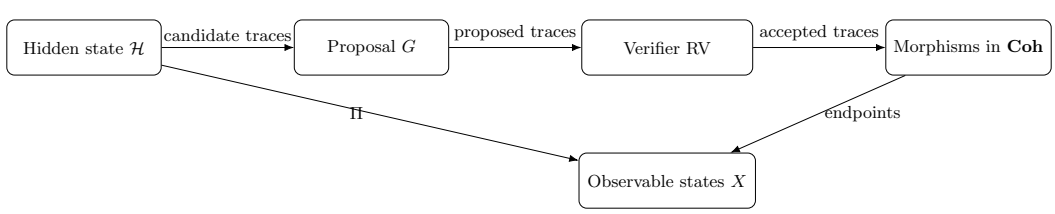

Figure 2. Coh execution wedge. Candidate traces enter a deterministic verifier; only certified traces become morphisms in the Coh category.

Figure 3 illustrates the separation between syntactic cost accounting and deeper semantic cost. V1 uses syntactic bookkeeping only. Semantic projection is reserved for V2, but the figure signals why V1 keeps spend and defect explicitly separated.

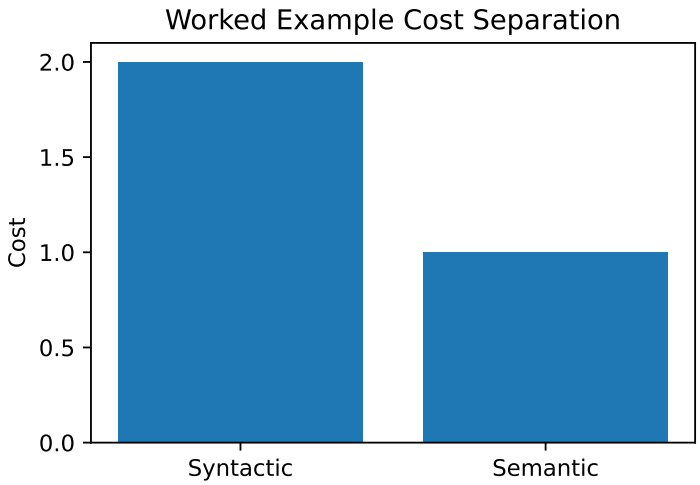

Figure 3. Cost separation. V1 uses exact syntactic bookkeeping; later projection-aware layers introduce semantic envelopes.

9 Failure Analysis

The category theorem is only as strong as its hypotheses. The following failures are structural, not cosmetic.

9.1 Verifier closure failure

If accepted traces are not closed under compatible concatenation, the hom-sets are not closed under composition. In that case the structure may form a partial category, a category with restricted composition, or a graph of certified transitions, but it is not the V1 Coh category defined here.

9.2 Non-additive bookkeeping

If spend or defect is not additive under concatenation, the resource proof in Lemma 4.4 breaks. A weaker theorem may still hold under subadditive or superadditive bookkeeping, but the inequality must be restated. The V1 theorem uses exact additive bookkeeping.

9.3 Infinite or non-cancellable valuations

The proof of closure cancels the intermediate valuation $V(y)$. If valuations are allowed to be $+\infty$, cancellation is not valid without extra domain restrictions. This is why V1 uses finite exact rational valuations.

9.4 Spend-defect collapse

If spend and defect are collapsed into a single scalar, the verifier loses the distinction between paid cost and tolerated uncertainty. The resulting category may still exist, but it no longer represents the Coh resource law as stated.

10 Lean and Implementation Alignment

The V1 construction is directly type-oriented. A Lean implementation can mirror the paper with the following records and theorem spine.

Lean-ready dependency spine.

`State` → `Step` → `Trace` → `RVAccept` → `Admissible` → `CohMor` → `Category`.

A minimal implementation separates data from proof obligations:

(1) states as a type X;
(2) valuation $V : X \to \mathbb{Q}_{\geq 0}$;
(3) traces as lists of typed steps with endpoint compatibility proofs;
(4) spend and defect as folds over trace steps;
(5) verifier acceptance as a predicate or deterministic Boolean with correctness theorem;
(6) admissible morphisms as records carrying verifier and inequality proofs;
(7) category laws as theorems over trace concatenation.

The critical implementation boundary is deterministic arithmetic. For production verification, numerical values should be serialized canonically, represented exactly, and compared without floating-point ambiguity.

11 V1-to-V2 Boundary

V1 proves that certified operational traces form a category. V2 adds hidden realizations, projection, and semantic envelope defect. The bridge is summarized in Table 1.

Table 1. Boundary between the V1 operational category and V2 projection-aware envelope.

Layer	V1 concept	V2 extension
Trace	finite accepted trace τ	observable trace R
Cost	$\mathrm{Spend}(\tau), \mathrm{Defect}(\tau)$	fiber envelope $\delta(R)$
Hidden data	not used	hidden realizations ξ
Projection	not used	$\Pi(\xi) = R$
Composition	concatenation $*$	observable composition $\odot$
Main theorem	category theorem	projection-induced subadditivity

Remark 11.1. The projection diagram in Figure 4 is not required for the V1 proof. It is included only to show the direction of the V2 upgrade: observable traces may correspond to multiple hidden realizations, so later reasoning must use worst-case fiber envelopes.

12 Conclusion

The V1 Coh category is a category of earned morphisms. A transition does not become an arrow merely because its endpoints exist. It becomes an arrow only after verifier acceptance and satisfaction of the governing resource inequality.

The main proved result is precise:

$$\begin{tikzcd}
\Theta_1 \arrow[dr, "\text{projection}"] & \\
& \tau \\
\Theta_2 \arrow[ur, "\text{projection}"'] &
\end{tikzcd}$$

multiple hidden realizations can collapse to one observable trace

Figure 4. Projection-aware extension. In V2, multiple hidden realizations may collapse to the same observable trace. V1 deliberately avoids this layer.

If the verifier accepts identities, accepted traces are closed under concatenation, and spend and defect bookkeeping are additive, then Coh-admissible traces form a category.

The trace algebra then gives a global resource law:

$$V(x_N) + \sum_{i=0}^{N-1} S_i \leq V(x_0) + \sum_{i=0}^{N-1} D_i.$$

Thus local admissibility composes into global resource control. This is the operational foundation on which the projection-aware V2 theory, metric enrichment, and governed dynamics layers can be built.

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